7. Monday, April 14th. More on quantifiers (2.4)

Last time: Existential and universal quantifiers and statements

## Statements with multiple quantifiers

**Example**: Write the next statement in symbols:

"For every natural number $x$, there exists a natural number $y$ such that $x<y$"

$$(\forall x \in \mathbb{N})(\exists y \in \mathbb{N})\ (x<y)$$

This is true: given $x \in \mathbb{N}$, we have $x+1 \in \mathbb{N}$ which satisfies $x < x+1$.

- Write the following in English:

$$(\exists y \in \mathbb{N})(\forall x \in \mathbb{N})\ (x<y)$$

"There exists a natural $y$ such that for every natural $x$ we have $x<y$".

This is false; there is no natural number larger than all naturals.

- Are these statements the same? No!

So order of quantifiers matters.

# Negating Quantified Statements

Example: Negate the statement "Everybody loves math".

$(\forall x \in U)$ ( x loves math) Where U = set of all people.

Incorrect answer: "Everybody does not love math"

$(\forall x \in U)$ (x does not love math)

Correct answer: "There exists some person who does not love math"

$(\exists x \in U)$ (x does not love math)

- Negation of Universal statement: $\neg (\forall x \in U)(P(x)) \equiv (\exists x \in U)(\neg P(x))$

Example: Negation of "At least one person loves hiking" $(\exists x \in U)$ (x loves hiking)

is "No one loves hiking" $(\forall x \in U)$ (x does not love hiking)

- Negation of Existential statement: $\neg (\exists x \in U)(P(x)) \equiv (\forall x \in U)(\neg P(x))$

**Example**: Negate "For every real number $x$, There exists a real number $y$ such That $x > y$"

Answer: In symbols: $(\forall x \in \mathbb{R})(\exists y \in \mathbb{R})(x > y)$

To negate:

$$\neg(\forall x \in \mathbb{R})(\exists y \in \mathbb{R})(x > y) \equiv (\exists x \in \mathbb{R})(\neg(\exists y \in \mathbb{R})(x > y))$$

$$\equiv (\exists x \in \mathbb{R})(\forall y \in \mathbb{R})(\neg(x > y))$$

$$\equiv (\exists x \in \mathbb{R})(\forall y \in \mathbb{R})(x \leq y)$$

"There exists some $x \in \mathbb{R}$ such That for every $y \in \mathbb{R}$ we have $x \leq y$".

# Direct Proofs (3.1-ish)

**Technique**: In a **direct proof**, we prove a conditional $P \longrightarrow Q$ by

1) **Assume** $P$ is True
2) **Create** a sequence of simple logical deductions using Things That we know
3) **Deduce** $Q$ is True

**Q**: What can we use To **create**?

1) Basic definitions (eg sets, even/odd integers, prime integers)
2) Statements we have already proven.
3) **Axioms**: Properties of The real numbers That we assume without proof.

# Axioms of The real numbers:

For us, The real numbers form a set where we can perform Two binary operations

- Addition (denoted $a+b$)
- Multiplication (denoted $a \cdot b$) \cdot

Satisfying The following properties:

Field Axioms (properties of addition and multiplication)

1) Closure: if $a,b \in \mathbb{R}$, Then $a+b \in \mathbb{R}$ and $a \cdot b \in \mathbb{R}$.

2) Commutativity: if $a,b \in \mathbb{R}$ Then $a+b = b+a$ and $ab = ba$

3) Associativity: if $a,b,c \in \mathbb{R}$ Then $(a+b)+c = a+(b+c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

4) Distributivity: if $a,b,c \in \mathbb{R}$, Then $a \cdot (b+c) = ab + ac$ (and $(a+b) \cdot c = ac + bc$)

5) Existence of identities: There are Two different real numbers 0 and 1 such That for every $a \in \mathbb{R}$ we have $a+0 = a$ and $a \cdot 1 = a$.

6) Additive inverses: For every $a \in \mathbb{R}$, There exists $-a \in \mathbb{R}$ such That $a + (-a) = 0$

7) Multiplicative inverses: For every $a \in \mathbb{R}$ s.t $a \neq 0$, There exists $a^{-1} \in \mathbb{R}$ such That $a \cdot a^{-1} = 1$

**Notation:** For $a, b \in \mathbb{R}$ we denote $a - b = a + (-b)$

If $b \neq 0$, we denote $\frac{a}{b} = ab^{-1}$

**Remark:**

1) The naturals $\mathbb{N}$ satisfy axioms (1)–(5)

2) The integers $\mathbb{Z}$ satisfy axioms (1)–(6)

3) The rationals $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z},\ b \neq 0 \right\}$ satisfy (1)–(7)

**Equality Axioms:** (For any notion of equality)

8) Reflexivity: For every $a \in \mathbb{R}$, we have $a = a$.

9) Symmetry: For any $a, b \in \mathbb{R}$, if $a = b$ Then $b = a$.

10) Transitivity: For all $a, b, c \in \mathbb{R}$, if $a = b$ and $b = c$, Then $a = c$.

11) Substitution: For all $a, b \in \mathbb{R}$, if $a = b$ Then $b$ may be substituted for $a$ in any mathematical statement without changing The Truth value of That statement.

AT This point, we may only assume definitions and axioms 1-11 for our proofs!

**Theorem 1:** Let $a, b, c \in \mathbb{R}$. If $\overbrace{a = b}^{P}$, Then $\overbrace{ac = bc}^{Q}$ and $a + c = b + c$.

**Proof:** We fix $a, b, c \in \mathbb{R}$ and suppose That $a = b$.

By closure of $\mathbb{R}$, we know That $ac \in \mathbb{R}$

By reflexivity of equality $ac = ac$.

Then, by The substitution axiom, we may substitute $b$ for $a$ on The right-hand side of The equality To deduce That $ac = bc$.

Similarly, by closure, $a + c \in \mathbb{R}$. Now since $a = b$, by substitution we get $a + c = b + c$

Hence $ac = bc$ and $a + c = b + c$, as desired. □

- Complete English sentences, properly punctuated.
- Each step is justified and follows order and direction.
- Various versions of □ To flag end of ~~proof~~.

**Theorem 2:** If $a,b,c \in \mathbb{R}$ are such that $a+c=b+c$, Then $a=b$.

Seems obvious but needs a proof. Not true for multiplication!

**Proof:** Suppose $a,b,c$ are such that $a+c=b+c$. By existence of additive inverse, we have $-c \in \mathbb{R}$ such that $c+(-c)=0$. Now, since $a+c=b+c$, by our previous Theorem, we have

$$(*)\quad (a+c)+(-c) = (b+c)+(-c)$$

Using associativity, we get

$$(a+c)+(-c) = a+(c+(-c)) \quad \text{and} \quad (b+c)+(-c) = b+(c+(-c))$$

So we can substitute in $(*)$ to get

$$a+(c+(-c)) = b+(c+(-c))$$

Since $c+(-c)=0$, we can substitute in both sides to get

$$a+0 = b+0$$

But, by the additive identity axiom, $a+0=a$ and $b+0=b$. So by substitution,

$$a = b.$$

As desired $\square$